When Failure Is Not Difficulty: Screening Reference–Physics Misalignment and Testing Adaptive Allocation on Exact Support for Humanoid Motion Tracking

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Abstract— **Persistent tracking failure need not identify useful practice: a retargeted reference may demand support unavailable under the declared robot and scene. We study this reference–physics misalignment (RPM) through model-relative screening and controlled allocation experiments. In a three-seed Unitree G1 campaign, top-1 exposure peaks at 87–89%, an unsupported kneel-and-crawl reference repeatedly attracts practice, and uniform sampling yields higher held-out survival than that sampler in each seed. A final-trajectory contact-capacity screen flags 2,442 of 10,705 AMASS-derived references in one retargeting pipeline and 7 of 4,950 in a separate production pairing; two implementations agree on 39 of 40 enriched-panel decisions. Feasibility features improve cross-policy difficulty transfer from Spearman correlation 0.567 to 0.609 over 100 held-out clips. We then bind training to 1,184 exact temporal units and compare four allocators at equal support and 49.2 million transitions per policy. Relative-progress allocation changes exposure but does not establish the registered tracking benefit: feasible-hard R−U TrackingScore is −0.0151, with paired seed-level 95% interval $[-0.0944, +0.0642]$. The broad non-regression criterion also fails, and exploratory learning curves favor uniform. These results separate reference admission, sampling intervention, and policy utility; they establish neither allocation equivalence nor a general advantage of gating.**

I. Introduction

Large retargeted motion banks have made generalist humanoid tracking a practical training recipe: map human motion to a robot, train a policy over the resulting references in massively parallel simulation, and spend additional updates on motions the policy currently fails. Systems such as PHC, MaskedMimic, H2O, ExBody, BeyondMimic, and SONIC demonstrate the scale of this recipe [1], [2], [3], [4], [5], [6]. Failure, completion, or recent learning progress then appears to identify where the next policy update is most useful [5], [7], [8].

That interpretation assumes every reference defines a solvable control problem. A retargeted trajectory may instead have no admissible contact capable of supplying its demanded base wrench, or may require joint effort beyond the modeled actuator range. We call this *reference–physics misalignment* (RPM). Under RPM, failures can reflect reference inadmissibility as well as limited experience or control capability. Our implemented screen addresses the first distinction. A coverage floor cannot create a behavior absent from the bank, and a progress estimate does not identify intrinsic difficulty or forgetting. The experimental question is therefore: given fixed reference support and a training budget, does learner-dependent allocation produce better tracking on the same held-out objective?

We observed the consequence in a 100-motion Unitree G1 campaign. A BeyondMimic-style sampler concentrated on the same kneel-and-crawl reference in all three seeds. Campaign peak top-1 mass was 0.884, 0.870, and 0.893; another clip owns the maximum in two seeds, but the same kneel-and-crawl reference is the recurrent attractor. It passes ordinary kinematic checks. Yet during its 0.75–1.75 s descent, the nearest collision geometry remains 7–10 cm above the floor while a median 329 N of support demand—approximately the 327 N model weight—has no modeled source.

CLIMB makes reference admission and practice allocation separately testable (Fig. 1). The `refeas` screen evaluates final robot-space references under a declared model; temporal units enforce support throughout each trial. A bounded allocator may use policy outcomes to change exposure within that support. Whether this feedback improves control is an empirical question, not a property of the interface.

Our contributions are:

1) an RPM diagnosis and model-relative contact-capacity screen, evaluated at bank scale and across implementations, with feasibility features that transfer difficulty information between policies sharing a learner;
2) an exact temporal-support interface and a matched four-arm, three-seed allocation study with a complete, inconclusive policy result and explicit exposure measurements.

We also report a bounded repair alternative: 22/26 screened candidates qualify, yet a one-policy paired deployment comparison does not establish an aggregate tracking gain. Admission's learning value requires a separate on/off experiment under a common allocator; the present allocation comparison cannot answer it.

II. Related Work

A. Motion filtering and dynamic retargeting

Policy-based curation already removes poorly tracked references. H2O retains AMASS motions that a privileged imitator can follow, and ExBody2 uses an initial policy's sequence errors to select a feasible and diverse subset [3],

[9]. These scores combine reference quality with policy capability. KungfuBot instead uses a human-space center-of-mass/center-of-pressure heuristic before robot retargeting [10]. LIMMT combines target-robot motion heuristics, diversity, and complexity, with physical-score weights calibrated through repeated policy training [11]. PHUMA filters source artifacts and retargets with joint-limit, ground-contact, and anti-skating losses [12]. CLIMB builds on this line but changes the training interface: the final robot-space screen determines the support on which an adaptive allocator is permitted to act.

Optimization-based retargeters address a complementary problem. Kinodynamic Motion Retargeting uses rigid-body and contact constraints to construct viable locomotion, while Direct Dynamic Retargeting uses simulator-in-the-loop optimization without an infeasible geometric intermediate [13], [14]. AMO and SPIDER likewise generate viable references through trajectory optimization or physics-based sampling with contact guidance [15], [16]. GMR further shows that retargeting artifacts affect downstream tracking robustness [17]. An earlier analytic study estimates velocity, acceleration, and torque requirements for 40 upper-body motions, but does not test whole-body environmental support [18]. Repair can therefore be preferable to exclusion. Our narrower question is diagnostic: after retargeting, can the demanded robot-space wrench be supplied by admissible contacts within modeled actuator limits?

B. Adaptive allocation and evaluation

Prioritized experience and level replay formalize the benefit and bias of nonuniform training distributions [19], [20]. In humanoid tracking, BeyondMimic uses a failure-EMA bin sampler, while GMT and EGM reweight motions or segments using tracking outcomes [5], [7], [8]. Those systems combine allocation with curation, staging, clipping, or architectural changes. Empirical failure, completion, and tracking error identify hard examples, but do not by themselves determine whether a reference is dynamically admissible. CLIMB inserts that test before outcome-based allocation; its matched comparison then changes only allocation over a shared set of legal trials. Learning-progress curricula already include ALP-GMM and portable implementations such as Syllabus [21], [22]. The contribution is the temporal-support contract and controlled test of policy utility, not normalization of a progress score alone.

Robot-learning comparisons are sensitive to seed count, aggregation, and interval construction [23]. Tracking adds a conditioning problem: kinematic error among survivors can improve when a method terminates earlier. HumanTracker similarly finds that kinematic metrics can miss support and contact failures [24]. We therefore use a liveness-weighted primary score and report common-survivor quality only beside survival.

III. Screening and Exact Temporal Support

A. Model-relative feasibility

Let a retargeted reference provide generalized position and velocity $(q_t, \dot{q}_t)$ for the G1 at frame t. We apply a centered five-frame moving average to velocity, differentiate at the reference frame rate, and smooth acceleration with the same window. Feasibility is defined relative to a robot model, scene, collision geometry, friction, actuator ranges, and reference. The present instantiation uses the G1 and a flat plane; electrical, thermal, compliance, latency, and structural limits remain outside this model-relative label.

We use the compiled MuJoCo model underlying the tracking environment. Contact-free inverse dynamics returns the generalized force required for the prescribed acceleration. The six free-joint coordinates define the base wrench $\mathbf{W}_t$ that the environment must supply; the remaining coordinates define contact-free joint torque $\boldsymbol{\tau}_{\text{free},t}$. A collision geometry becomes a candidate contact when its exact distance to the plane is at most $g = 0.06\,\text{m}$.

The 6 cm band is a declared tolerance. At 3 cm a known feasible control is flagged on 42% of its frames because that clip's stance frames sit approximately 3 cm above the plane, a residual we attribute to retarget ground alignment but have not measured bank-wide. At 10 cm the attractor's 7–10 cm floating descent is incorrectly granted a contact candidate. The chosen setting lies between these observed failure modes; it is not a universal constant.

B. `refeas`: contact-capacity screen

For each candidate point we construct a four-edge pyramidal friction cone with coefficient 0.6 in the primary model; a simulator-matched view makes non-foot contacts frictionless. Mapping nonnegative generator magnitudes f_t through the contact Jacobian yields both base-wrench matrix $\mathbf{A}_t \mathbf{f}_t$ and joint-torque contribution $\mathbf{J}_t \mathbf{f}_t$. A weighted nonnegative least-squares problem first exposes the unconstrained residual, with angular-wrench rows weighted by two. We then solve a linear program minimizing ℓ_1 wrench slack subject to

$$-\boldsymbol{\tau}_{\max} \le \boldsymbol{\tau}_{\text{free},t} - \mathbf{J}_t \mathbf{f}_t \le \boldsymbol{\tau}_{\max}, \qquad \mathbf{f}_t \ge 0. \tag{1}$$

The translational component of the resulting weighted residual defines unsupported force. With no candidate contact, the residual is the contact-free demand.

We report the fraction of frames with no contact candidate (`airborne_frac`), the fraction whose torque-limited unsupported force exceeds half the robot weight (`infeasible_frac`), normalized unsupported-force integral, and joint-torque infeasibility. A clip crosses the primary rule only when `infeasible_frac`> 0.10 (strict inequality). A frame whose torque-limited program is itself infeasible in contact has no finite unsupported force and enters `infeasible_frac` as zero; such frames are reported separately as joint-torque infeasibility, which is nonzero for 734 clips. Counting them as unsupported would flag 2,502 clips (23.4%) rather than 2,442, so the reported rate is the conservative convention. Ballistic flight is not flagged merely for being airborne: near-gravity acceleration has little external support demand. Conversely, nonballistic hovering or descent without support is flagged.

An independent implementation screens the 4,950-clip production bank in 179.8 s of wall time on eight workers, or 0.29 CPU-seconds per clip over 1,818,423 screened frames (0.79 ms per frame). The screen therefore runs as an offline bank-ingestion stage rather than in the policy control loop.

C. Repair as a routing alternative

DFRP applies a bounded contact projection to qualified floating-reference cases. It lowers and smooths root clearance, then adjusts supporting-leg joints with damped contact inverse kinematics and joint-limit projection. The changed trajectory is re-screened and must satisfy residual infeasibility at most 5%, root displacement at most 8 cm, valid joint limits, contact-IK residual at most 10 mm, and at least one legal 50-step start. Missing-scene cases bypass projection. This is an implemented routing option, not a guarantee of usable control: repair changes the tracking target and needs its own paired policy evaluation.

D. Exact-support gated allocation

Clip classification is too coarse because one reference can contain both supportable and unsupported intervals. We reduce per-frame screens to ordered feasible runs, retain only 50-step windows wholly inside a run, and bind each motion and sidecar by SHA-256. Across 800 training clips, 1,841 source runs yield 1,184 admissible units and 368,951 legal starts after 657 short runs are discarded. A unit is an attribution stratum; the deployment prior is uniform over legal starts, not units or clips.

For an admitted interval $\mathcal{A}_u$, define legal starts by every reference frame accessed during horizon H and lookahead configuration $\mathcal{L}$:

$$\mathcal{S}_u = \{s : s + \mathcal{K}(H, \mathcal{L}) \subseteq \mathcal{A}_u\}. \tag{2}$$

The current interface includes the terminal observation and uses current-frame reference observations. "Exact" denotes membership relative to this screen and access contract; it is not a certificate of physical feasibility.

Let b_u be the normalized legal-start prior over admitted units and $g_u = |s_u(k) - s_u(k-10)|$ the nonnegative change of Beta-smoothed conditional success at sampler clock k. Uniform U uses b. Absolute-progress A retains the earlier fixed additive floor. Relative-progress R, before active caps and for complete history with $\mu = \sum_u b_u g_u > 0$, is exactly

$$p_u = 0.8 b_u + 0.2 \frac{b_u g_u}{\mu}. \tag{3}$$

The implemented prior/cap fallback covers incomplete history or zero μ. Common positive rescaling of g leaves Eq. (3) unchanged, and its pre-cap total variation from b is at most 0.2. These are exposure properties, not guarantees of useful practice. The sampler clock advances every 50 environment steps, so the ten-tick window spans 500 steps, and conditional rates use a Beta prior with per-tick decay 0.99. Equation (3) is the general rule at exploration ratio 0.4 and relative factor 2. All arms share a 10-tick progress window and per-unit and per-clip ceilings of 0.05 and 0.25. A ranks the same absolute progress but adds a fixed floor of 0.05 before a 0.4 exploration mixture; R replaces that floor with the scale-relative form of Eq. (3). D ranks conditional failure at exploration 0.8 with difficulty power 1, so R−D compares allocator designs rather than isolating ranking alone.

Intervals do not all receive the same remedy. Admissible runs and DFRP-qualified repairs enter the exact support; missing-context and residual failures do not. This routing preserves feasible subsequences rather than treating every screen flag as a whole-clip deletion instruction. The four-arm comparison uses the original admitted bank; it substitutes no repaired targets.

A trial fails when a non-timeout termination fires: anchor height error above 0.25 m, anchor gravity-projection error above 0.8, or height error above 0.25 m at an ankle or a wrist. Truncation at a segment boundary is not a failure. Every arm adds the same one-off terminal cost of -10 on failure to the inherited tracking rewards. The runtime samples only legal starts, emits explicit truncation at a segment boundary, never wraps into a rejected frame, and serializes its deterministic generator and sufficient statistics. Per-unit and per-clip caps bound concentration. Missing sidecars, changed motion hashes, invalid starts, invalid frames, or censored resets fail the contract.

The paired evaluator freezes the motion, start, replicate, environment-noise seed, and dynamics-noise seed across policies. It assigns the reference before the first observation, disables auto-reset until terminal channels are captured, rejects invalid offsets, and records checkpoint, task, condition, reference, evaluator, and output hashes. The 100-clip panel is name- and hash-disjoint from training and contains 2,800 paired conditions.

IV. Experimental Design

A. E1: RPM attractor diagnosis

We analyze the previously sealed 100-motion campaign: 4,096 environments, 4,000 PPO iterations, three seeds per arm, and 100 disjoint evaluation motions with eight episodes per clip. It includes failure-adaptive, clip-uniform, and normalized grounded samplers. We report exposure, attractor identity, held-out survival, and the attractor's modeled contact-capacity residual. This campaign establishes the motivating failure case; the later four-arm study separately compares exact-support allocators. The historical adaptive arm adds exploration mass before normalization, which does not guarantee a probability floor. Its collapse therefore diagnoses that particular sampler/reference combination. The current D baseline instead retains a normalized 80% prior component; the historical result does not establish that every failure-adaptive allocator will collapse under RPM.

B. E2: bank scale, agreement, and transfer

The primary corpus contains 10,705 AMASS-derived references retargeted through `whole_body_tracking` to

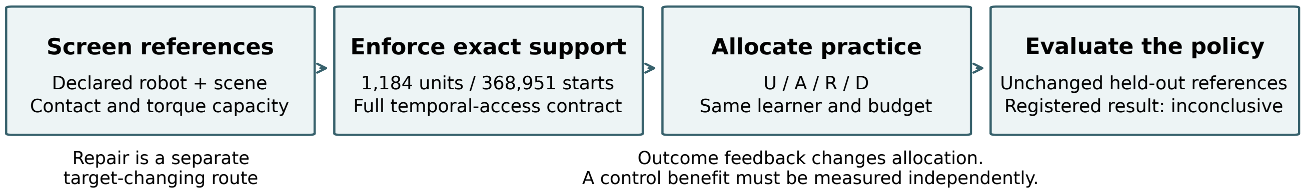

Fig. 1. **Admission, allocation and policy utility are separate tests.** The screen supplies model-relative support; the temporal interface enforces legal reference access; independent evaluation measures the resulting policy. The tested outcome-to-allocation feedback changed exposure but did not establish the registered tracking benefit. Neither screen qualification nor exact support implies hardware feasibility. Repair is a separate, target-changing route.

G1. We report raw counts under the fixed rule and category/source breakdowns. A separate experiment covers all 4,950 references in the filtered BONES-SEED production pairing. We never pool their prevalence. A deterministic 40-clip panel, 20 from each bank and enriched for flags, passes through both implementations to measure score and decision agreement; it is not a prevalence sample. Separately, a ridge model fits 11 intrinsic reference features to per-clip difficulty under one curriculum and ranks difficulty under another on the same 100 held-out clips. We compare intrinsic features with and without the three `refeas` features against 200 control designs that append three i.i.d. standard-normal columns in their place; this is cross-policy transfer, not cross-architecture transfer.

C. E3: DFRP repair qualification

We freeze a source- and severity-stratified CPU panel before repair: 26 strict-flagged candidates span three root-displacement bands and two initial-infeasibility bands, and four source-matched feasible clips serve as no-op controls. The primary outcome is exact-ready flagged count under the 5% residual, 8 cm root, joint-limit, 10 mm contact-IK, source-hash, and legal-start gates. The gate requires at least 20/26 candidates, 4/4 byte-identical ready controls, zero integrity or qualification failures among admitted clips, and a reproducible manifest payload. Per-clip CPU runtime is a secondary diagnostic.

D. Four-arm allocation on unchanged exact support

An earlier two-arm E4 protocol selected absolute-progress settings using allocation-only calibration. Its long-run manipulation check subsequently failed, leaving policy endpoints closed. A separate prospective confirmation retained A as a declared comparator and added relative progress R and conditional failure D alongside uniform U.

The completed study uses seeds 21, 22 and 23, 512 environments, 4,000 PPO iterations and 24 rollout steps: 49,152,000 simulator transitions per policy. Support, learner, rewards and reference targets are common. All twelve training gates passed before any held-out cell opened. Checkpoints 1000, 2000, 3000 and 3999 give 48 evaluation cells on 100 unchanged held-out clips, including 25 feasible-hard clips. That stratum is fixed before training: panel clips are ranked by the mean percentile rank of seven reference quantities (peak required friction, peak vertical force, contact-switch rate, flight fraction, non-foot ground contact, peak angular momentum, and negated mean support margin), and the top 25 are taken. No policy outcome enters the rule. A cell's 2,800 conditions are 100 clips $\times$ 7 phase starts $\times$ 4 replicates, each an independent episode in its own environment, never concatenated. Both error terms are means over the survived prefix of an episode, evaluated over uninterrupted windows of up to three seconds.

The actor and critic are observation-normalized ELU multilayer perceptrons with hidden widths 512/256/128, taking 160 and 286 observation coordinates, respectively; the actor outputs 29 joint-position commands. Physics runs at 200 Hz with 50 Hz control. PPO uses Adam (initial learning rate 10^{-3}, adaptive KL target 0.01), five epochs, four minibatches, clipping 0.2, discount 0.99, GAE coefficient 0.95 and entropy coefficient 0.005. These settings are shared across arms.

The primary score is liveness-weighted reference tracking:

$$h \exp\left(-\frac{\mathrm{MPKPE}}{0.30\,\mathrm{m}} - \frac{\text{anchor error}}{0.40\,\mathrm{rad}}\right), \tag{4}$$

where h is survived-horizon fraction, MPKPE is the mean position error of the 14 tracked bodies expressed relative to the `torso_link` anchor, and anchor error is that anchor body's orientation error. We average conditions within clip, clips within panel and paired differences across training seeds. Independent replication is the trained seed pair, not the number of conditions.

The frozen primary decision requires final hard R−U mean at least +0.02, a positive lower two-sided 95% paired seed t bound (df=2), and all-panel lower bound above −0.01. A pass would establish a positive effect with a target-reaching point estimate, not a true effect of at least +0.02. Paired hierarchical bootstrap intervals are supplementary. Learning curves, normalized area under the observed curve (AULC), and attainment of the paired U final score are exploratory; non-attainment is retained. No seeds are appended to the completed decision.

V. Results

A. E1: Curriculum collapse under RPM

Adaptive peak top-1 mass is 0.884/0.870/0.893 across campaign seeds, and the same BMLmovi kneel-and-crawl clip repeatedly becomes dominant in all three. The maximum belongs to another clip in two seeds. Uniform sampling achieves higher held-out endpoint survival in every paired seed: differences (uniform minus adaptive) are +0.0300, +0.0275, and +0.0300. With only three seeds, the sign pattern is the useful description; the minimum attainable paired permutation p-value is 0.125. The campaign's third arm separates the sampler defect from outcome-driven allocation as such. A grounded sampler, a convex mixture holding exactly 10% of the mass on the uniform prior, peaks at top-1 mass 0.568/0.649/0.696 instead of 0.87–0.89, and its endpoint survival exceeds uniform in all three paired seeds (+0.0088, +0.0238, +0.0125; mean +0.015). Collapse therefore tracks the additive-floor construction rather than outcome-driven allocation itself. At three seeds this ordering is descriptive, and it does not license a grounded-sampler recommendation.

The reference violates no joint limit in the model's declared ranges. The dynamic screen instead localizes the failure. From 0.75–1.75 s, 86% of frames have no candidate contact and median unsupported force is 329 N; the model weighs 327 N. The 1.75–7.25 s kneeling interval is supportable through non-foot contacts, and the 8.0–8.5 s rise again becomes unsupported. Under a whole-clip start with a 10 s horizon no trained policy survives the reference, stopping after 2.38 s on average. Under three-second stratified windows the picture is protocol-dependent: a frame-zero window survives in 0.25 to 1.00 of episodes across seven evaluated policies, while an 8 s start completes the remaining 1.96 s in all seven. We therefore report the horizon with every survival number and do not read the late-offset result as evidence about the descent.

B. E2: Bank-scale screening and difficulty transfer

Under the strict rule, 2,442 of 10,705 primary-pipeline clips are flagged (22.8%). Category rates range from 12.9% for quiet motion to 58.6% for dynamic motion, and source rates from 0.1% for GRAB to 100% for CNRS. Severity-stratified inspection shows why labels are insufficient: five inspected CNRS cases are ordinary fast walks with extended floating intervals, while the Transitions sample mixes ingest defects with acrobatic content.

Only 7 of 4,950 production clips (0.14%) cross the same nominal rule, while 111 (2.24%) are airborne on more than 10% of frames. Five of seven flags are jumps, four naming a 50 cm box absent from the declared flat scene. Their verdict is *unsupported on this plane*, routing them toward missing terrain or exclusion rather than geometric root projection.

Across the stratified 40-clip panel, `infeasible_frac` rank correlation is 0.9836, `airborne_frac` correlation is 0.9974, and strict decisions agree for 39/40 clips (Cohen's $\kappa = 0.9485$). The disagreement is `burpee_002__A362_M` (0.019 versus 0.136). Figure 2 keeps the full-bank denominators separate and uses only this same-input panel to check implementation agreement; it does not establish corpus or retargeter equivalence.

TABLE I
REPAIR QUALIFICATION AND DEPLOYMENT ANSWER DIFFERENT QUESTIONS.

Evidence / status	Result and scope
Qualification / measured	22/26 flagged candidates; 4/4 feasible controls byte-identical.
Deployment / exploratory	One policy, 26 clips (22 qualified repairs plus 4 byte-identical controls), 656 paired conditions per arm: raw 0.3925, repaired 0.3915; difference −0.0010, clip-bootstrap 95% CI $[-0.0086, +0.0080]$. On the 22 repaired clips alone the difference is −0.0015 $[-0.0105, +0.0090]$, while the unchanged controls return +0.0020 $[-0.0025, +0.0085]$, which bounds evaluator reproducibility at twice the headline magnitude.

The difficulty decomposition is also visible across policies trained with different curriculum allocations. On the same 100 held-out clips, an 11-feature intrinsic atlas fit to adaptive-policy difficulty ranks uniform-policy difficulty at $\rho = 0.567$. Adding `infeasible_frac`, `airborne_frac`, and normalized unsupported impulse raises the rank correlation to 0.609, above 198 of 200 control designs in which three i.i.d. standard-normal columns replace the `refeas` features (one-sided $p = 0.010$). This control establishes that the gain is not attributable to added design width alone; it is not a comparison against three other real reference features. All six directed policy pairs move in the same direction and four have $p < 0.05$; the two pairs targeting the grounded policy do not. This is transfer across policies sharing an architecture, not evidence of cross-architecture or cross-robot transfer.

C. E3: Contact-manifold repair via DFRP

DFRP qualifies 22/26 flagged candidates (84.6%) and all four feasible controls pass as byte-identical no-ops. The resulting curated view, the 22 qualified repairs plus the 4 controls, contains 36 admissible units and 10,561 legal 50-step starts. Across all 30 panel clips, median and 95th-percentile CPU runtime are 2.57 and 6.29 s per clip. Two candidates remain above the 5% residual-infeasibility ceiling, and two more reach that ceiling but violate the 10 mm contact-IK bound; all four are quarantined. Thus 84.6% is a qualification rate on the frozen panel, not a bank-wide recovery rate or a policy-benefit estimate.

The fixed-policy deployment result (Table I) does not establish an aggregate gain, safety, or equivalence. It compares tracking of raw and repaired targets; it is not a retraining effect or improved tracking of the unchanged original reference.

D. Completed allocation result: inconclusive

The earlier E4 absolute-progress arm reaches mean post-warm-up TV 0.0297, below its 0.05 gate. Its disposition is `not_tested`; it supplies no policy null. The newer confirmation retains A under its own comparator rule. R changes exposure in all three seeds (mean TV 0.08348/0.08249/0.08233), as does D

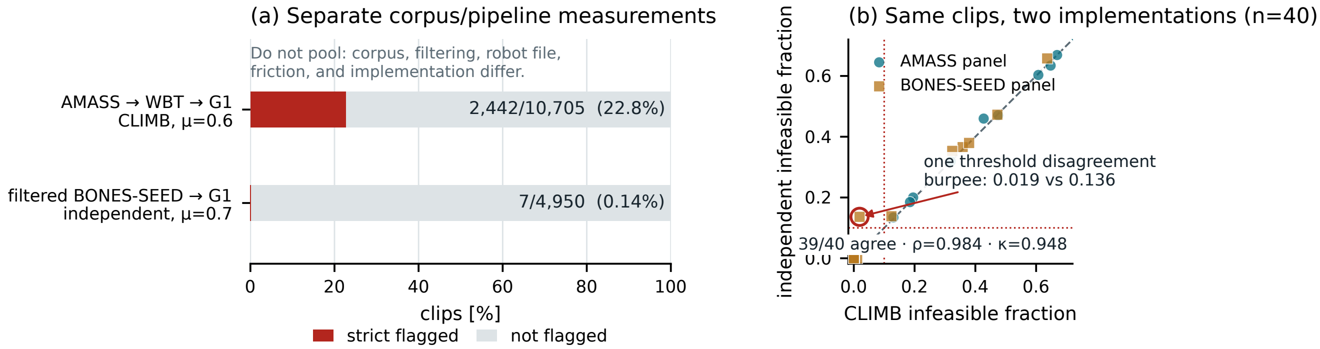

Fig. 2. Bank-scale screen and implementation agreement. (a) Bars are normalized within two separate corpus/pipeline denominators and apply the strict `infeasible_frac`>0.10 rule: 2,442/10,705 for AMASS→WBT→G1 and 7/4,950 for filtered BONES-SEED→G1. Corpus, filtering, robot file, friction, and implementation differ, so this is not a causal retargeter comparison. (b) On a flag-enriched, deterministic 20+20 same-clip panel, strict decisions agree for 39/40 clips (Spearman $\rho = 0.9836$, Cohen's $\kappa = 0.9485$). That panel checks implementation agreement; its stratified selection is not a prevalence sample.

TABLE II

Final TrackingScore differences. Three paired trained seeds; two-sided seed t intervals, df=2.

R−U	Feasible-hard	All-panel
Seed 21	−0.03400	−0.04892
Seed 22	−0.03299	−0.01402
Seed 23	+0.02178	+0.03500
Mean	−0.01507	−0.00932
95% lower	−0.09436	−0.11405
95% upper	+0.06422	+0.09542

TABLE III

Earlier controls, with their original scope and status.

Study / status	Bounded finding
E-HYG / sealed	Whole-clip pruning: feasible-panel survival 0.918 → 0.907; $\Delta = -0.0101$; one seed pair, permutation over held-out feasible clips, $p = 0.951$.
Soft weighting / failed gate	Hard-rejected mass 0.199; this does not implement exact admission.
E4 / not tested	Mean TV 0.0297 below 0.05; policy endpoint unopened.
Repair-all / exploratory	Deployment contrast +0.0397 misses +0.05 target and coverage gate.

(0.08391/0.08587/0.08398), while A stays near uniform at 0.02976/0.02934/0.02982 and therefore acts as a weak-separation comparator rather than a second strong intervention. A learning-free reference for this functional form, with 1,184 units, equal prior mass, inactive caps and independent Gaussian progress changes, yields mean total variation 0.0605 over 2,000 replicates and exceeds 0.05 in every replicate. Exposure change above the gate therefore records that allocation moved, not that it tracked learning progress. The reference is an illustration under idealized assumptions and is not subtracted from the observed values. All twelve training runs pass their declared gates with zero recorded invalid or censored events. Across 492 saved states, exact sampler and checkpoint replay establishes intervention integrity.

The frozen primary result is *inconclusive* (Table II). Neither the primary rule nor the broad non-regression guard passes. Two hard-panel seeds favor U and one favors R. The interval spans both meaningful harm and benefit; it does not establish either, nor does failure of the guard prove regression. The supplementary paired bootstrap hard interval is $[-0.04417, +0.02532]$ and cannot override this decision. Descriptive final hard R−D is -0.00675 with interval $[-0.02189, +0.00840]$; R−A is -0.00668 with interval $[-0.07243, +0.05907]$. These contrasts do not establish equivalence or progress-specific superiority.

Exploratory normalized hard AULC R−U is negative in all three seeds: -0.03978, -0.03042, -0.01676 (mean -0.02898; descriptive 95% t interval $[-0.05774, -0.00023]$). Two R seeds never attain their paired U final score on the observed grid. These data do not support an efficiency-accelerator claim; sparse checkpoints do not identify a precise crossover. The secondary interval is not a new confirmatory harm test. Figure 3 reports the complete curves.

a) Earlier controls remain separate: Table III separates prior protocol statuses. None is pooled with the completed four-arm result or substituted for a matched admission on/off test. Segmentation retained 12.5 of the 20.2 minutes in 99 flagged training clips and lost only three whole clips at zero guard. Retained duration is not demonstrated retained skill. A matched on/off admission comparison under D is the next distinct test; no incomplete result enters this manuscript.

VI. Limitations

CLIMB's admission label is model-relative. The current screen instantiates one G1 model and a flat scene with collision contacts, friction, and actuator force limits. Hardware deployment would add electrical, thermal, compliance, latency, and structural constraints and then require closed-loop validation. Terrain-bearing references likewise require their terrain in the scene; the four box-jump flags in the production bank are routed to missing context rather than

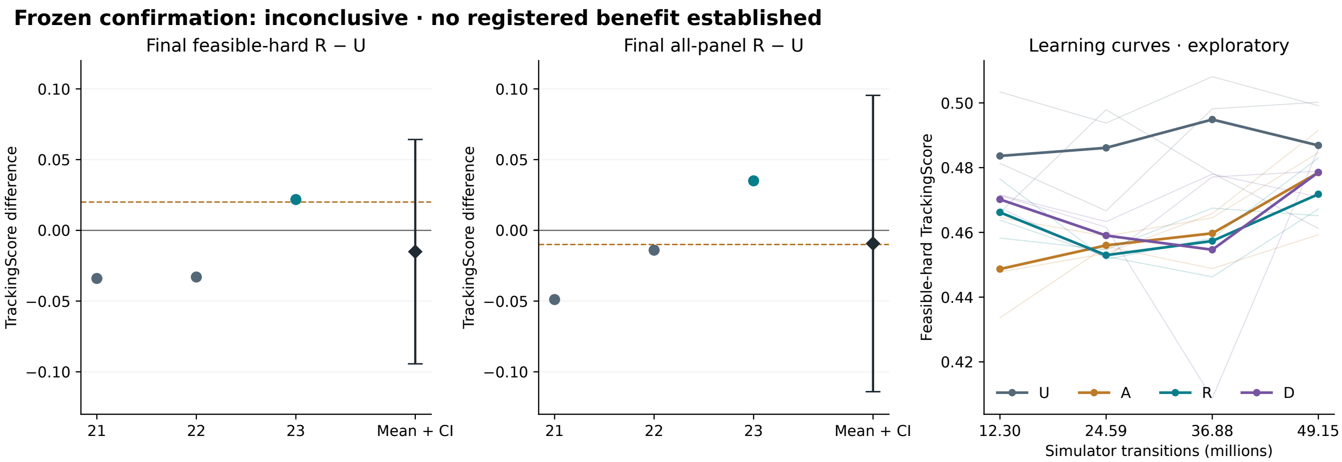

Fig. 3. Completed allocation confirmation. Seed pairs, means and two-sided 95% t intervals appear at left and center. Dashed lines are the hard point-estimate target and all-panel lower-bound margin. Thin learning curves are seeds, thick curves are means; the cost axis is simulator transitions. Curves and attainment are exploratory.

declared bad motion.

Finite differences, the 6 cm contact band, and the half-weight residual threshold introduce modeling choices. The sensitivity evidence is local: two clips across three contact bands and one clip across three residual thresholds. Bank-scale sensitivity of either threshold is untested, and a new robot or retargeter must recalibrate them. The 22.8% and 0.14% rates therefore remain separate corpus–pipeline measurements. A smooth feasibility weight derived from residual slack could retain borderline cases, but would couple admissibility and allocation; it requires its own matched ablation against the current binary gate.

The allocation study has only three independent seed pairs. Its observed hard-panel standard deviation 0.03192 yields a 95% t half-width 0.07929, much larger than the +0.02 point target. This is an observed precision limitation, not a retrospective power guarantee or evidence of equivalence. Thousands of evaluation conditions cannot replace independent training. The current result neither establishes benefit nor rules out a target-sized benefit; a future replication requires its own fixed sample size.

The current simulator clamps knee and hip-roll effort at ±139 N m and adds zero command delay on plane terrain; physical G1 transfer is untested. Our configuration audit records vendor-published knee maxima of 90 and 120 N m for two G1 revisions, but we archive no snapshot of that source and cannot confirm that those figures and the compiled limit describe the same point in the drivetrain, so we treat this as a recorded discrepancy rather than a measurement. The screen compiles the same actuator ranges as training. Re-screening the 900-clip bank behind our exact support at 120 and 90 N m leaves 888 clips unchanged and moves the flagged count from 99 to 98 of 900, with no clip newly flagged, so on this bank the screen's decisions are dominated by the unsupported-wrench residual rather than by the actuator channel. We had predicted that a tighter limit could only raise the flagged fraction, and measurement refuted it: the reported quantity is the translational component of a residual whose total the program minimizes, so tightening can redistribute slack between components, and four clips fall by at most 0.024. We therefore make no lower-bound claim. The actor additionally consumes two terms that the upstream hardware-oriented G1 configuration removes, base linear velocity and the reference anchor pose in the torso frame, so a physical deployment would have to rebuild the observation interface and not only the dynamics. Numerical parity remains unresolved in a separate physical-sensitivity development check, so its perturbation scores are not reported as evidence. A continuous adapter traverses 8.58-second training references, but comparative full-motion execution is untested. The frozen-policy instrumentation pilot does not establish a noise floor; its planned 17.6-million-transition burn-in makes it a separate study. Practice interventions, reliability-calibrated progress and hardware evaluation remain future work, not explanations established by the current allocation result.

VII. Conclusion

RPM makes persistent failure an ambiguous instruction for practice. The diagnosed campaign, bank-scale screen and cross-policy difficulty transfer motivate explicit measurement of reference admissibility. Exact temporal support then permits a controlled allocation test: relative progress changes exposure, but the completed three-seed comparison does not establish its registered tracking benefit. A qualified repair likewise does not establish a fixed-policy gain. Reference qualification, exposure and control utility therefore need separate evidence. The learning value of admission under a common allocator remains the next direct experiment.

Reproducibility and Acknowledgments

The stack uses mjlab v1.6.0 at commit `0fb8a681136b` and MuJoCo 3.11.0. Motion, sidecar, unit-table, task, checkpoint, evaluator, condition and output identities are SHA-256 bound. Licensed motion payloads cannot be redistributed; release requires reconstruction instructions and permitted

aggregate metadata. OpenAI Codex and Anthropic Claude assisted with code, analysis tooling, figures and manuscript drafting in Sections I–VII; all claims are tied to verified artifacts and their recorded provenance.

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